7 Figure 7.1 shows a battery of e.m.f. 6.5 V connected in series with a motor, a fixed resistor and a switch.

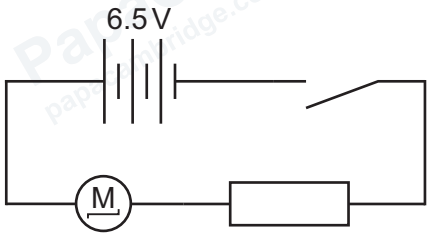

Figure 7.1

(a) Complete the equation that defines e.m.f. E.

E =

[1]

(b) The switch is closed. The battery moves 4300 C around the circuit in 1.8 h.

(i) Calculate the work done by the battery.

work done = ... J [2]

(ii) Complete the equation that defines electric current I.

I =

[1]

(iii) Show that the current in the motor is 0.66 A.

[1]

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(iv) The resistance of the fixed resistor is 2.5 Ω.

Calculate the p.d. across the motor.

p.d. across motor = ... V [3]

[Total: 8]

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